

CertChain

Digital certificate issuance and public proof verification

Assignment report | 6 October 2026

EVIDENCE DRAFT - Hosted issue, revocation, proof, and PDF/QR verified; email, expiry, and video pending.

1. System Overview

CertChain lets an educational organization create and manage digital certificates while a public visitor checks whether an issued certificate has a matching blockchain proof. An organization administrator signs in, creates a draft, confirms issuance, downloads the generated PDF, and can later revoke the certificate with a private reason. A recipient or verifier opens the QR link or enters CERT-YYYY-NNNNNN on the public page.

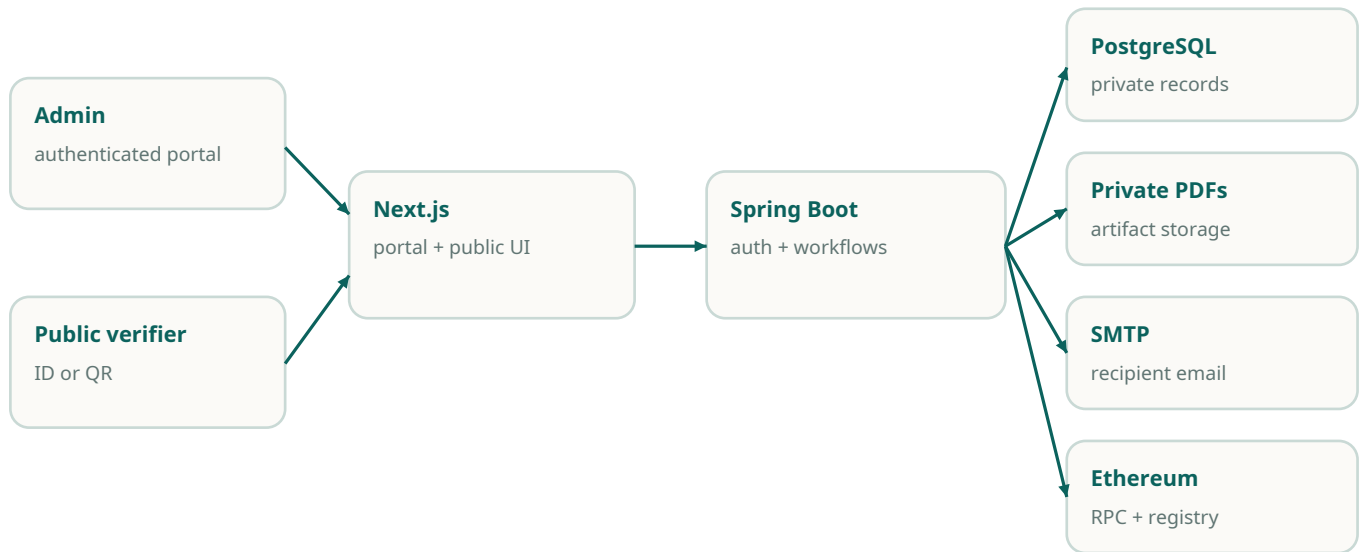
The design separates three kinds of information. PostgreSQL is the operational database for organizations, users, certificate details, recipient email, transaction and delivery journals, and private revocation reasons. Private file or object storage holds generated PDFs. Ethereum stores only a deterministic certificate key, a bytes32 SHA-256 proof, issue and optional expiry timestamps, issuer address, and a one-way revocation flag. The chain does not contain names, emails, PDF bytes, or the revocation reason.

The implemented system runs locally with PostgreSQL and Hardhat. A free-tier deployment uses Vercel Hobby, Render Free, Neon Free PostgreSQL and private storage, and a source-verified contract on Sepolia. One hosted certificate remains valid, and a second was issued then revoked; both public results matched their confirmed Sepolia proof. Live email, hosted expiry evidence, and the public video remain pending; the deployment record is included in this package.

2. System Architecture

The repository has a Next.js/React frontend, a Java 21 Spring Boot 4.1.1 backend, a PostgreSQL database with Flyway migrations, and a Solidity `CertificateRegistry` contract. The frontend renders the portal and public verification pages. The backend owns authentication, tenant authorization, canonicalization, signing, transaction recovery, PDF generation, and email delivery. It accesses the chain through web3j and a server-side issuer wallet.

Figure - Architecture flow



The frontend can suggest valid inputs and guard navigation, but the backend authorizes every protected route using the verified JWT's organization ID. The portal never sends an authoritative hash, status, transaction hash, role, or issuer private key.

3. User Flow / System Flow

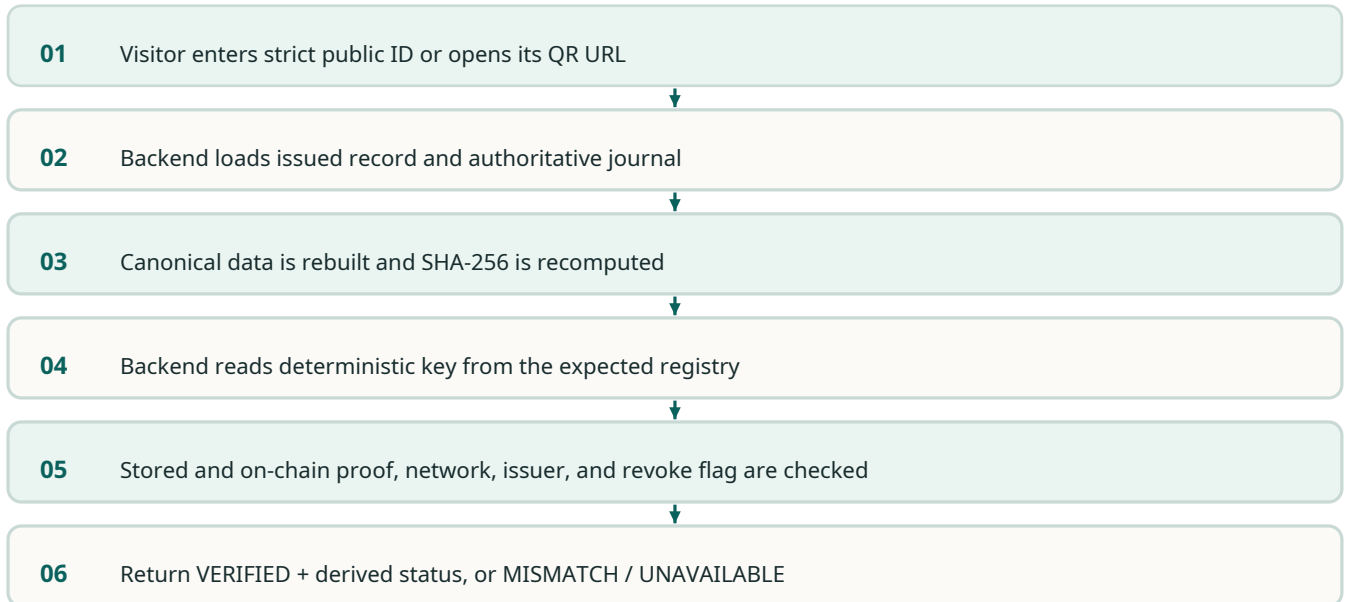
Issuance. An admin creates a tenant-owned draft. The server allocates the public ID with a PostgreSQL global yearly sequence. On confirmation, the backend locks the record, freezes proof-relevant fields, computes the canonical SHA-256 hash, and commits a CREATED blockchain journal before sending a transaction. It stores the returned hash as SUBMITTED, then marks the certificate ISSUED only after validating the receipt, confirmations, expected contract, issuer, event, and on-chain record. A timeout stays pending for reconciliation; a failed receipt cannot become issued. PDF generation and email run afterward and can be retried without another chain issue call.

Figure - Issuance flow



Verification. The public route accepts only a strict public ID and exposes issued certificates. The backend rebuilds the canonical string, recomputes SHA-256, compares it with the stored hash and chain proof, checks the expected network/contract and revocation consistency, then derives status. Unknown, malformed, and draft IDs share a not-found response. An RPC outage or proof mismatch does not produce a false verified or valid result.

Figure - Verification flow



Revocation. The admin enters a private reason and explicitly confirms. The backend journals CREATED and SUBMITTED, validates a `CertificateRevoked` receipt/event, and only then sets `revokedAt`. A timeout stays pending for recovery, and an uncertain transaction is not blindly resent. Public status gives REVOKED priority over EXPIRED; expiration is derived at read time after the UTC expiry date.

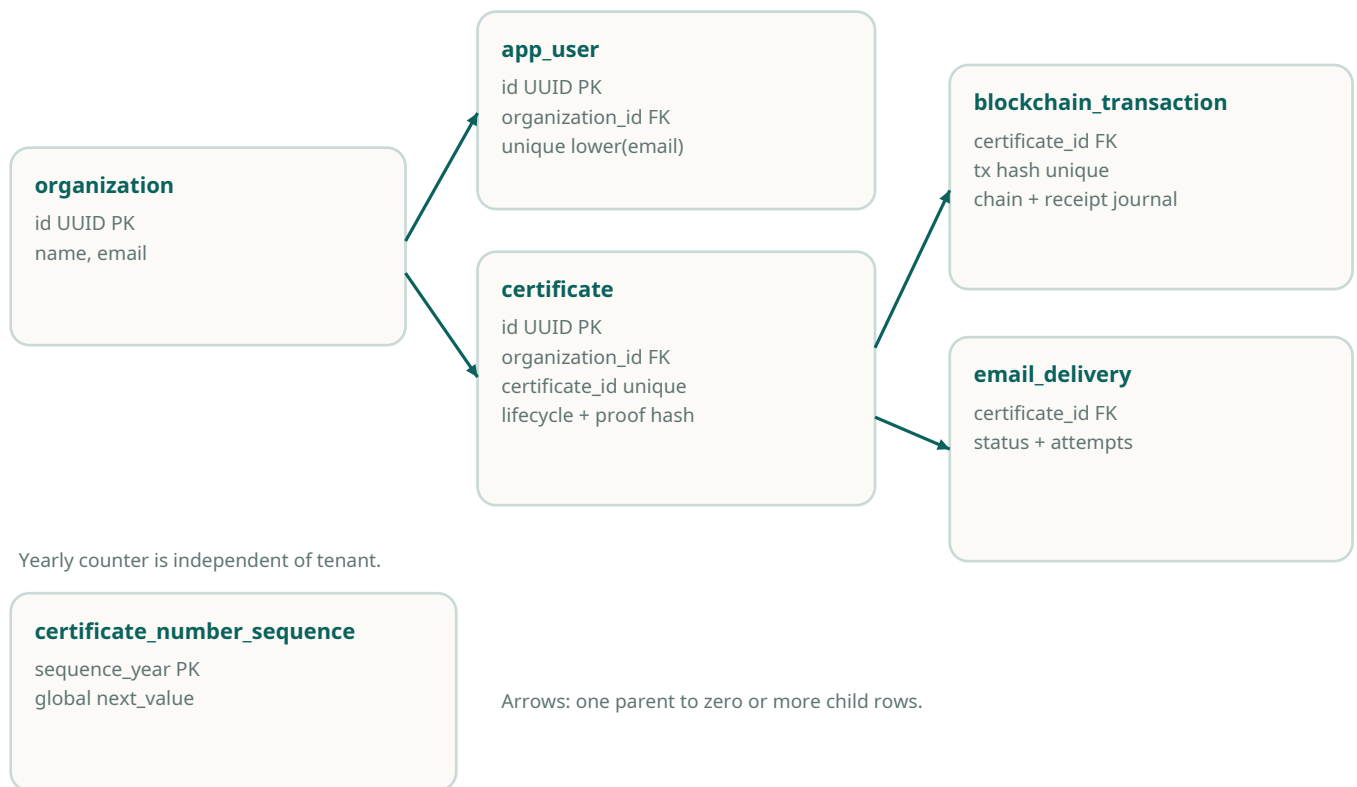
Figure - Revocation flow



4. Database Design / ER Diagram

Flyway migrations create normalized tables for organization, app_user, certificate, blockchain_transaction, email_delivery, and certificate_number_sequence. A certificate belongs to one organization, has zero or more transaction and email journal rows, and retains its own immutable public ID. Transaction hash, chain ID, network, contract, block, confirmation, and failure details live in blockchain_transaction, not duplicated on certificate. Public VALID/EXPIRED/REVOKED status is computed, not persisted.

Figure - ER flow



UUIDs are internal keys; certificate_id is unique and formatted CERT-YYYY-NNNNNN. The global yearly sequence uses one atomic PostgreSQL upsert per allocation, allows gaps after rollback, and rejects exhaustion after 999999. The database also enforces case-insensitive unique user emails, nonnegative journal values, valid enum strings, transaction-hash uniqueness, and expiry_date >= issue_date when expiry exists. Foreign keys restrict deletion of issued history.

5. Blockchain Architecture

The backend computes `certificateKey = keccak256(UTF-8(uppercase(trim(public ID))))`, which gives a stable lookup key. Anyone who knows or guesses the public ID can compute this key; hashing does not make the ID confidential. The proof is `SHA-256(UTF-8(canonical data))`. Version v1 fixes the field order: public ID, recipient name, program, organization UUID, issue date, expiry date. Text is Unicode NFC normalized, trimmed, internal whitespace collapsed, and backslash/pipe/equals escaped. The ID is uppercased with `Locale.ROOT`; names and programs keep their case. Dates use ISO format and a missing expiry is empty. This prevents logically equivalent text from producing accidental different proofs.

The backend signs transactions with a server-held issuer key stored in Render's secret environment; the user browser never gets this key. A separate admin wallet controls contract roles. The [Sepolia registry](#) is source-verified.

[Hosted issuance `0x3e690ae4...19c6690`](#) succeeded in block 11852629 with a decoded `CertificateIssued` event. A separate [hosted revocation `0x180f2dad...f856e028`](#) succeeded in block 11852907 with a decoded `CertificateRevoked` event. Transaction journals preserve `CREATED`, `SUBMITTED`, `CONFIRMED`, and `FAILED` states across process failures. Reconciliation checks a known hash or on-chain event/key before a retry, avoiding duplicate issuance or revocation.

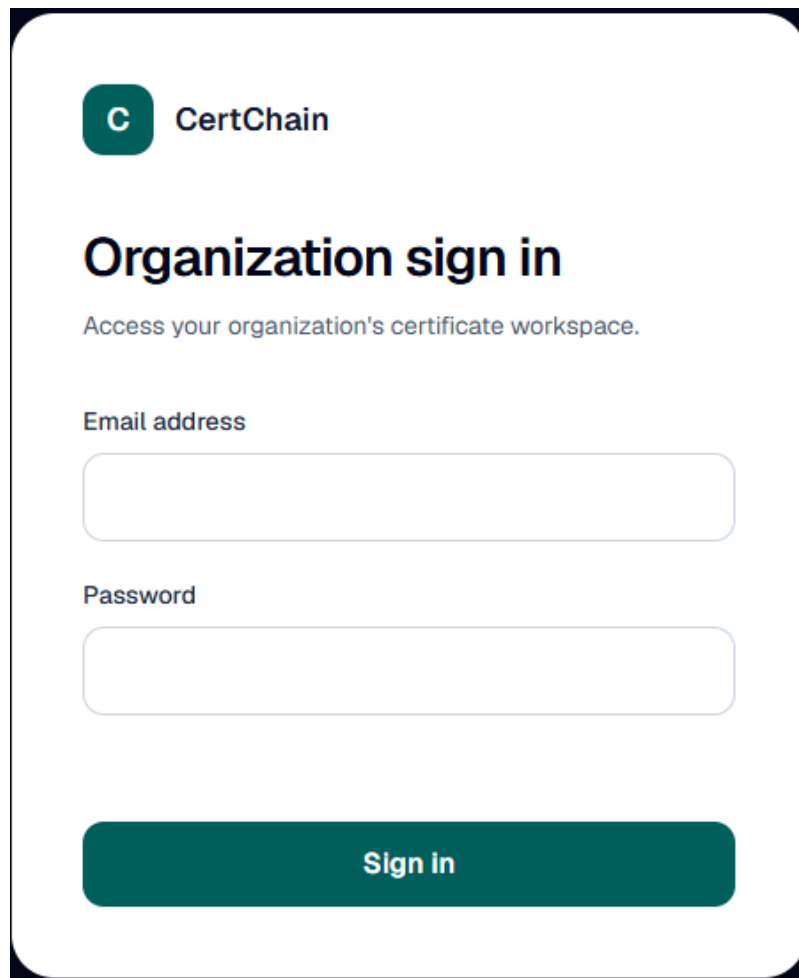
6. Smart Contract Design

`CertificateRegistry.sol` uses `OpenZeppelin AccessControlDefaultAdminRules` with an explicit admin, a one-day delayed admin transfer, and `ISSUER_ROLE` for issue and revoke. The constructor rejects zero addresses. Issuance rejects zero key/hash, duplicates, and expiry at or before the block time. Records are never overwritten. Revocation rejects missing or already revoked keys. Indexed `CertificateIssued` and `CertificateRevoked` events support receipt validation and recovery. Public `read` and `verify` functions return deterministic results; on-chain expiry is true only after `expiresAt`, and application status prioritizes revocation.

The contract stores only key, proof hash, issue/expiry timestamps, issuer, and revoked flag. Eight contract tests passed locally on 6 October; coverage reported 100% line and statement coverage for `CertificateRegistry.sol`. Etherscan displays an Exact Match source verification for the deployed Sepolia address.

7. User Interface Design / Screenshots

The portal uses a restrained educational trust style with clear labels, focus states, loading/empty/error states, and text/icon status cues. The screenshots below are actual browser or PDF renders. Local verification states came from a synthetic PostgreSQL and Hardhat demonstration; hosted captures use the real Vercel/Sepolia deployment.



The image shows a login interface for CertChain. At the top left is the CertChain logo, which consists of a dark green square with a white letter 'C' inside, followed by the text 'CertChain' in a dark grey sans-serif font. Below the logo is the heading 'Organization sign in' in a large, bold, dark grey font. Underneath the heading is a subtext 'Access your organization's certificate workspace.' in a smaller, lighter grey font. The form contains two input fields: 'Email address' and 'Password', both with light grey labels and rounded rectangular input boxes. At the bottom of the form is a large, dark green button with the text 'Sign in' in white, bold, sans-serif font.

Local login screen

 **CertChain**

Administrator [Sign out](#)

[Dashboard](#) [Certificates](#) [New certificate](#) [Organization profile](#)

ORGANIZATION PORTAL

Dashboard

A current view of your certificate activity.

+ Issue certificate

Issued certificates



0

Confirmed on chain

Currently valid



0

Issued and active

Expired



0

Past expiry date

Revoked



0

Permanently revoked

Hosted organization dashboard after admin login



Administrator

[Sign out](#)[Dashboard](#)[Certificates](#)[New certificate](#)[Organization profile](#)[← Certificates](#)

CERTIFICATES

Create draft

Review the details before issuing. The public ID is generated when you save.

Draft created

Public ID: CERT-2026-000001

[View certificate](#)

Hosted draft creation with server-issued public certificate ID

Draft details

The public ID stays the same when you edit this draft.

[Edit draft](#)

Proof and issuance

○ DRAFT

Blockchain status is currently unavailable.

Transaction
Not submitted

Transaction state
Not started

Block
Pending

Confirmed at
Pending

Issue certificate

Certificate

Recipient
CertChain Demo Recipient

Deployment Smoke Test

Issue date
2026-10-05

Expiry date
No expiry

Hosted certificate draft details before issuance

CertChain

Sign out

Dashboard

Certificates

New certificate

Organization profile

← All certificates

CERTIFICATE DETAILS

CERT-2026-000001

CertChain Demo

ISSUED ✓ Valid

Proof and issuance✓ Issued

Proof confirmed on chain.

Transaction
0x3e690ae42fda4f90066c922bc3310e4b2384160cd47c98645fd
b1f15819c6690

Transaction state
CONFIRMED

Block
11852629

Confirmed at
2026-10-06T01:30:00Z

[View transaction in explorer](#) [View public verification](#)

Certificate PDF

Ready to download

Download PDF

Recipient email

Email delivery is disabled on this server.

Revocation

Confirm revocation to submit the transaction.

Review revocation

Certificate

Recipient
CertChain Demo Recipient

Program
Deployment Smoke Test

Issue date
2026-10-05

Expiry date
No expiry

Hosted certificate issued with confirmed Sepolia transaction

Free-tier deployment verified - remaining gates and video pending

Page 10

[← Search another certificate](#)

Certificate verification

**Valid**

The certificate data matches the confirmed blockchain proof.

Certificate ID

CERT-2026-000001

Organization

CertChain Demo

Recipient

CertChain Demo Recipient

Program

Deployment Smoke Test

Issue date

2026-10-05

Expiry date

No expiry

Network

sepolia (chain 11155111)

Transaction

0x3e690ae42fda4f90066c922bc3310e4b2384160cd47c98645fdb1f15819c6690

Contract

0x293597447c1e39825Df82fc3791BA7AB3Eb9719f

Block

11852629

Block timestamp

2026-10-06T01:30:00Z

Copy ID

Copy transaction

View transaction in explorer

Hosted public verification shows valid proof

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Etherscan

HomeBlockchainTokensNFTsAPIMore

Transaction Details

Sponsored: Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. Build with AI.

Overview

Logs (1)

State

Transaction Receipt Event Logs

184

Address

0x293597447c1e39825df82fc3791ba7ab3eb9719f

Name

CertificateIssued (index_topic_1 bytes32 certificateKey, index_topic_2 bytes32 certificateHash, index_topic_3 address issuer, uint64 issuedAt, uint64 expiresAt) View Source

Topics

0

0xe1e68a6175f44ef852dfec1ca2ccb8322ccc059c33b61fc40baf243f912d35

1: certificateKey

Dec

⇒ 9A0092E37CF8C54E3B19084C5781AC77C9ABD9FFBFD06E465BAA4D58E56FD3C2

2: certificateHash

Dec

⇒ 392C7CB2E4FFCE339809FB02919EF14C4D63722671593E2B406422EA177B83900

3: issuer

Dec

⇒ 0x2F28e790a7f6c93C1d471142D6ab5c600F42408d

Data

issuedAt (uint64) : 1791250200

expiresAt (uint64) : 0

Powered by Ethereum

Back to Top

Add Sepolia Test Network

This website uses cookies to improve your experience. By continuing to use this website, you agree to its Terms and Privacy Policy. Got It!

Etherscan © 2026 (Sepolia)

ations: 0x71c765...d8976f

Sepolia Etherscan transaction with decoded issue event

Free-tier deployment verified - remaining gates and video pending

Page 12

**CertChain Demo**

CERTCHAIN / VERIFIED CREDENTIAL

CERTIFICATE OF ACHIEVEMENT

PRESENTED TO

CertChain Demo Recipient

FOR COMPLETING

Deployment Smoke Test

ISSUED

5 October 2026

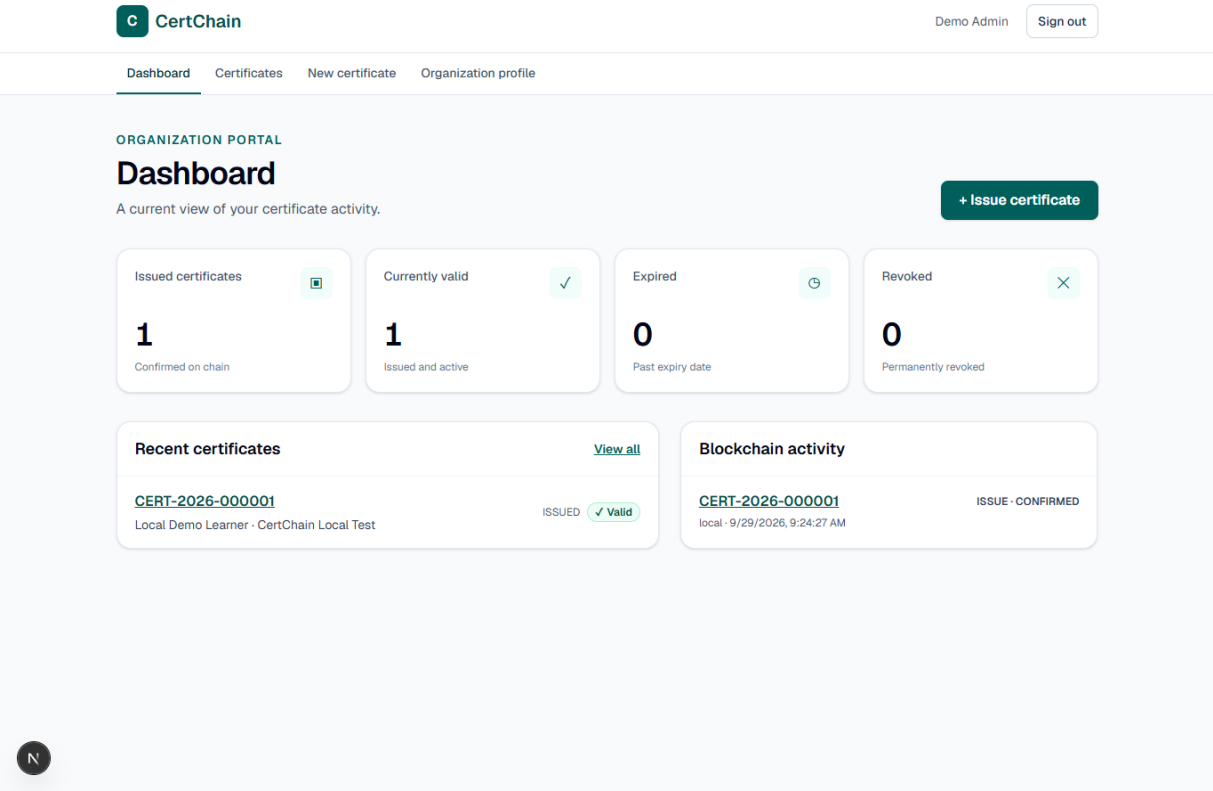
CERTIFICATE ID

CERT-2026-000001

Verify this certificate online:

<https://cert-chain-gold.vercel.app/verify/CERT-2026-000001>

Hosted certificate PDF rendered after confirmed Sepolia issuance; its QR was independently decoded



Local dashboard with actual API counts and confirmed issuance activity, captured 29 September 2026

[← Search another certificate](#)

Certificate verification

 **Valid**

The certificate data matches the confirmed blockchain proof.

Certificate ID	Organization
CERT-2026-000003	Phase 12 Demo Academy
Recipient	Program
Tamper Example	Proof Integrity Workshop
Issue date	Expiry date
2026-09-28	No expiry
Network	Transaction
local (chain 31337)	0x5898044348371ae161e3bf34df72636df6ba9c7645a dd3ca57f24cccb060666f
Contract	Block
0x5FbDB2315678afecb367f032d93F642f64180aa3	4
Block timestamp	
2026-09-28T16:52:45Z	

Copy ID

Copy transaction

Valid local-chain verification

[← Search another certificate](#)

Certificate verification



Expired

The proof matches, but the certificate expiry date has passed.

Certificate ID	Organization
CERT-2026-000004	Phase 12 Demo Academy
Recipient	Program
Expiry Example	Boundary Workshop
Issue date	Expiry date
2026-09-28	2026-09-28
Network	Transaction
local (chain 31337)	0xe325ed89a5e8cd27eaa05efa04fde12732897bb74f78cd3245eeb308836ff78d
Contract	Block
0x5FbDB2315678afecb367f032d93F642f64180aa3	5
Block timestamp	
2026-09-28T16:53:32Z	

Copy ID

Copy transaction

Expired local-chain verification

[← Search another certificate](#)

Certificate verification

 **Revoked**

The proof matches and the issuer has revoked this certificate.

Certificate ID

CERT-2026-000001

Recipient

Ada Example

Issue date

2026-09-28

Network

local (chain 31337)

Contract

0x5FbDB2315678afecb367f032d93F642f64180aa3

Block timestamp

2026-09-28T16:52:14Z

Organization

Phase 12 Demo Academy

Program

Secure Systems Workshop

Expiry date

No expiry

Transaction

0xe956ba1e2eaabb18b67fbd903e224fc7771539527f262791a862264cec102d6a

Block

3

Copy ID

Copy transaction

Revoked local-chain verification

[← Search another certificate](#)

Certificate verification



Not found

No issued certificate was found for this public ID.

Unknown certificate response

[← Search another certificate](#)

Certificate verification

 Proof mismatch

The stored certificate and blockchain proof do not agree. Do not rely on this certificate.

Check again

Certificate ID
CERT-2026-000003

Organization
Phase 12 Demo Academy

Recipient
Tamper Example

Program
Tampered Workshop

Issue date
2026-09-28

Expiry date
No expiry

Copy ID

Proof mismatch from synthetic tampering



PDF downloaded after confirmed local issuance; its QR was independently decoded to the displayed localhost URL

The local screenshots and PDF are synthetic captures from separate runs; repeated public IDs do not represent one certificate history. The local QR decoded to its localhost URL. The hosted PDF is included in this package; it was downloaded, visually checked, and its QR decoded exactly to the public URL for CERT - 2026 - 000001. The separate hosted [revoked result](#) and [explorer event](#) were verified on 6 October. A third [hosted fixture](#) expires after 6 October UTC and was still VALID that day; its actual EXPIRED result needs a next-day check. Saved hosted revoked/expired screenshots and real recipient email remain pending. Local chain hashes are not Sepolia proof.

8. Implementation Summary

Implemented features include authentication with BCrypt and short-lived JWT cookies, CSRF protection, tenant-safe draft management, deterministic proof hashing, a role-controlled registry, failure-safe issuance/revocation journals, public verification, PDF/QR generation, bounded email delivery, and a tenant dashboard. The backend stores authoritative chain metadata in transaction rows; PDF and email failures do not undo confirmed issuance. The hosted backend uses Neon PostgreSQL and private object storage. One confirmed Sepolia issuance remains valid, and another was successfully revoked with public proof. Live SMTP remains disabled pending Brevo phone verification and a real mailbox test.

On 6 October 2026, Docker was restored and two full backend runs each passed 73 tests with zero failures, errors, or skips, including PostgreSQL Testcontainers migration, constraints, security, tenant, workflow, and delivery tests. An additional Mailpit smoke test passed with a synthetic multipart email and PDF attachment. Frontend lint/build and all 31 tests passed. Contract typecheck, all eight tests, and coverage passed; `CertificateRegistry.sol` reported 100% line and statement coverage. The included dated validation record distinguishes these local results from hosted proof. The hosted PDF was inspected and its QR independently decoded. Hosted revocation and public re-verification passed. A complete hosted browser walkthrough, real mailbox delivery, the expiry fixture's next-day result, and backup/recovery check remain open.

9. Public GitHub Repository Link

<https://github.com/Thna17/CertChain> is the original public team repository; the free deployment currently builds from <https://github.com/Seypa-47/CertChain>. The [public app](#), [API health endpoint](#), [verified contract](#), and [hosted issuance transaction](#) are public. Their current availability must be checked again before submission.

10. Individual Contribution Report

Git history through f196d2a contains exactly two author identities. **Hong Than Brathna** (hangbrathna10@gmail.com) authored foundation commit 8b67919 on 22 September 2026. That commit established the repository, Next.js/Spring Boot/Hardhat scaffolds, Maven wrapper, Docker Compose, environment templates, initial interface and security skeleton, and first architecture, API, database, flow, blockchain, and checklist documents.

At the report evidence cutoff, **Seypa47** (khemrakupasey01@gmail.com) authored 22 subsequent commits through f196d2a. Their diffs added the database/domain layer, PostgreSQL tests, authentication, draft management, registry contract, hashing and blockchain workflows, verification, PDF/QR, revocation, email, dashboard, security tests, deployment preparation, free hosting integration, and report work. These statements describe Git-authored changes, not a measured percentage of effort. The distribution is visibly uneven: one foundation commit and 22 later commits. No PR, review, design-session, pair-programming, or separate test-session record was supplied. Commit count alone cannot establish a fair effort split. Only Git history is used for attribution, as requested; later submission-preparation commits are outside this dated tally.

11. Public Demo Video Link

Pending. The student plans to film the demo and has not provided a public URL. An ordered narration and recording checklist are included in this package. The report must be finalized only after a real URL is available and checked. The video should show the hosted Sepolia evidence and clearly label any local synthetic tamper or expiry fixture.